Assignment - 11

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45/50

Task-1:- linear Regression,

Given

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

a) find the regression equation

$$b_1 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

$b_1 = 1.99$, $b_0 = 0.05$

$$y = 0.05 + 1.99x$$

b) predict at $x = 6$

$$y = 0.05 + 1.99(6)$$

$$= 11.99.$$

c) calculate errors

RSS = 0.107, MSE = 0.0214, $R^2 = 0.9973$

d) Reliability

The R^2 value is very high, showing a good fit.

Task-2:- Ridge Regression

$S_{xx} = 10$, $S_{xy} = 19.9$, $\lambda = 5$

$$b = \frac{S_{xy}}{S_{xx} + \lambda}$$

$$= \frac{19.9}{10 + 5}$$

$$= 1.327$$

$WL_8 = \frac{}{10}$

percentage shrinkage:

$$\frac{1.99 - 1.327}{1.99} \times 100$$

$$= 33.33\%$$

Increasing λ increasing bias and decreas variance. It also reduces overfitting.

Task-3 :- Naive Bayes classifier

Given

spam = 5 emails, Ham = 5 emails, Total = 10 em

a) calculate prior probabilities.

$p(spam) = 5/10 = 0.5$

$p(Ham) = 5/10 = 0.5$

probability	spam	HAM
p(offer = Y)	4/5	1/5
p(offer = N)	1/5	4/5
p(link = Y)	4/5	1/5
p(link = N)	1/5	4/5

b) classify offer = Y, Link = Y

$$p(spam/x) = \frac{0.5(0.8)(0.8)}{0.5(0.8)(0.8) + 0.5(0.2)(0.2)}$$

$$= 0.9412$$

laplace smoothing ($\alpha = 1$)

$$P(\text{spam}|X) = \frac{0.5\,(5/7)(5/7)}{0.5\,(5/7)^2 + 0.5\,(2/7)^2}$$

$$= 0.8621$$

Offer = N, Win = N

$$P(\text{spam}|X) = \frac{0.5\,(2/7)^2}{0.5\,(2/7)^2 + 0.5\,(5/7)^2}$$

$$= 0.1379$$

Naive Bayes assumes features are independent

Task-4 :- logistic Regression

$w_1 = 0$, $w_2 = 0$, Bias = 0

learning = 0.5.

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$P_1 = P_2 = 0.5$$

Step-1 :- calculate gradients.

$$\frac{\partial L}{\partial w_1} = 0.25, \quad \frac{\partial L}{\partial w_2} = -0.25 \quad \frac{\partial L}{\partial b} = 0.$$

Step-2 :- update weights.

$$w_1 = 0 - (0.5)(0.25) = -0.125$$

$$w_2 = 0 - (0.5)(-0.25) = 0.125$$

$$b = 0$$

Step-3 :- New predicted probabilities

$$P_1 = 0.5312 \,, \quad P_2 = 0.4688$$

step-4 :- loss.

Before update : 0.6931

After update = 0.6326.

The loss decreases after updating the w

Task - 5 :- decision tree and information ga

Total samples = 10

yes = 6, NO = 4

step-1 :- entropy of parent node

$H = -0.6 \log_2(0.6) - 0.4 \log_2(0.4)$

$H = 0.971$

step-2 :- entropy

A = a → 4 yes, 0 NO → entropy = 0

A = a → 2 yes, 4 NO → entropy = 0.918

step-3 :- Information gain

IG = 0.971 - 0.551

= 0.420

step-4 :- Gini impurity

$1 - (0.6^2 + 0.4^2) = 0.48$

$= 0.6(0) + 0.4(0.444)$

$= 0.2667$

$0.48 - 0.2667 = 0.2133$

The split is worthwhile because both IG and

Gini decrease are positive

Task-6:- cost complexity pruning

Training samples = 100

subtree errors = 3

$R_\alpha(T) = R(T) + \alpha |leaves|$

$R(T) = 3/100 = 0.03$

$R(leaf) = 8/100 = 0.08$

step-1:- find critical α

$0.03 + 4\alpha = 0.08 + \alpha$

$3\alpha = 0.05$

$\alpha = 0.0167$

step-2:- Decision

α value	Decision
0.01	Keep subtree
0.02	prune subtree

A smaller α keeps the detailed tree.

Task-7:- Neural network & Backpropagation

Given

$x_1 = 1, \quad x_2 = 2$

$w_1 = 0.4, \quad w_2 = 0.3, \quad v = 0.5$

setp-1:- calculate hidden output

$h = \sigma(0.4(1) + 0.3(2))$

$h = \sigma(1) = 0.7311$

step-2:- calculate final output

$$\hat{y} = \sigma(0.5 \times 0.7311)$$

$$\hat{y} = 0.5904$$

step-3:- calculate error

$$E = \frac{1}{2}(1 - 0.5904)^2$$

$$= 0.0839$$

step-4:- Backpropagation

$$\delta_o = (\hat{y} - 1)\,\hat{y}\,(1 - \hat{y})$$

$$\delta_o = -0.0991$$

$$\delta_h = \delta\,(0.5)\,h\,(1 - h)$$

$$= -0.00974$$

step-5:- updated weight

$$w = 0.5362$$

$$w_1 = 0.4049$$

$$w_2 = 0.3097$$

$$E_{new} = 0.0811$$

Backpropagation updates the weights.

Task-8:- support vector machine (SVM)

$$w = (0.5, 0.5)\,a, \quad b = -2$$

$$f(x) = 0.5x_1 + 0.5x_2 - 2$$

a) verify support vectors

(3, 3)

$$f(3, 3) = 1$$

(1, 1)

$$f(1, 1) = -1.$$

Margin width

$$\frac{2}{\|w\|} = \frac{2}{\sqrt{0.5}}$$

$$2.828$$

b) classification and distance

(4, 2)

$$f(4,2) = 1$$

$$d = \frac{|1|}{0.7071} = 1.414$$

(1, 2)

$$f(1,2) = -0.5$$

$$d = \frac{0.5}{0.7071} = 0.707$$

c) Hinge loss

$$L = \max(0, 1 - y f(x))$$

(1, 2), $y = -1$

$$L = \max(0, 1 - 0.5) = 0.5$$

(2, 2), $y = +1$.

$$L = \max(0, 1 - 0) = 1$$

Both points violate the margin because their hinge losses are positive

Task-9:- Ensemble learning

classifier accuracy = 70%

a) Three classifiers

$$P = \binom{3}{2} (0.7)^2 (0.3) + 0.7^3$$

$$= 0.784 = 78.4\%$$

b) five classifiers

$$p = \sum_{k=3}^{5} \binom{5}{k} (0.7)^k (0.3)^{5-k}$$

$$p = 0.8369 = 83.69\%$$

3 c) Analysis

3 classifiers = 78.4%

5 classifiers = 83.69%

Accuracy increased when more independent classifiers are combined.

Task-10:- AdaBoost

Total samples = 5

Initial weight of each sample = 1/5 = 0.2

step-1:- Weighted error

$$\epsilon = 0.2 + 0.2 = 0.4$$

step-2:- learner weight

$$\alpha = \frac{1}{2} \ln\left(\frac{1-\epsilon}{\epsilon}\right)$$

$$= \frac{1}{2} \ln\left(\frac{0.6}{0.4}\right)$$

$$\alpha = 0.2027$$

step-3:- update weights

classified samples

$$0.2 e^{-0.2027} = 0.1633$$

misclassified samples

$$0.2 e^{0.2027} = 0.2449$$

Step-4:- Normalised weights

samples	updated weight
1	0.1667
2	0.8500
3	0.1667
4	0.1667
5	0.2500

Adaboost increases the weights of misclassified samples.

Task-(1):- Discriminant Analysis

$$\Sigma = \begin{pmatrix} 2 & 0.5 \\ 0.5 & 1 \end{pmatrix}$$

$\mu_1 = (1, 2)$, $\mu_2 = (3, 1)$

a) find discriminant direction

$$w = \Sigma^{-1}(\mu_2 - \mu_1)$$

$$\Sigma^{-1} = \begin{pmatrix} 0.5714 & -0.2857 \\ -0.2857 & 1.1429 \end{pmatrix}$$

$$w = \begin{pmatrix} 1.4286 \\ -1.7143 \end{pmatrix}$$

$$T = \frac{1}{2} w^T(\mu_1 + \mu_2)$$

$$T = 0.2857$$

b) classify $x = (2.5, 1)$

$$w^T x = 1.4286(2.5) - 1.7143(1) = 1.8571$$

1.8571 is greater than 0.2857

class P2

c) unequal priors

$p(w_1) = 0.8$ $p(w_2) = 0.2$

$T = 0.2857 + \ln(0.8/0.2)$

$= 1.6720$

1.8571 is greater than 1.6780

class P2

The decision remains unchanged even after changing the class priors.

Task 4:- classifier Evaluation

Total loans = 1000

FN cost = 50,000

Fp cost = 5000

a) calculate performance metrics

$$\text{Accuracy} = \frac{TP+TN}{\text{Total}} \qquad \text{precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{Tp+fN} \qquad F_1 = \frac{2PR}{P+R}$$

Metric	Model A	Model B	Model C
Accuracy	88%	80.5%	93%
Precision	43.75%	32.08%	66.67%
Recall	70%	85%	60%

b) calculate total expected cost

cost = (FN × 50000) + (FP × 5000)

Model-A:-

= (30 × 50000) + (90 × 5000)

= 19, 50, 000

Model-B:-

= (15 × 50000) + (180 × 5000)

= 16, 50, 000

Model-C:-

= (40 × 50000) + (30 × 5000)

= 21, 50, 000

Model	Total cost
A	19,50,000
B	16,50,000
C	21,50,000

c) Which model should be depolyed?

Model C has the highest accuracy and f1-score

However, Model B has the lowest cost. because it detect more defaulters.

d) Improvement methods

1. Tune the classification threshold to reduce false negative

2. use class-weighted learnings to give more importants to defaulters

3. use cross-validation to prevent overfitting.